Reference Event 13: The CenturyLink Outage, December 2018

Characterization Against the Communications Sector Design Basis

DBA-COMM-RE-13 — Reference Event Instrument, DBA-COMM Series

Lineage: Issued under the Communications Sector Framework v0.12; DBA-MA Parent Framework; conferred on the proposal of Operations and Coordination (DBA-COMM-OpsCo)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the outage of CenturyLink's fiber transport network that began on 27 December 2018, in which four malformed management packets generated by one switching module in Denver were rebroadcast from node to node across the network until the nodes could no longer carry traffic, and service, including 911 for some seventeen million customers in twenty-nine states, was disrupted for almost thirty-seven hours. It reads the outage back through the Communications Sector Design Basis, as the Communications Sector Framework states it and as DBA-COMM-OpsCo (*Operations and Coordination*) applies it to the coordination layer of the communications network, and asks which provisions of the class bore on it, how each performed, and what the class did and did not correctly draw from it. The Federal Communications Commission reported on the outage and settled its investigation, and the Washington Utilities and Transportation Commission penalized the carrier for its effect on 911 in that state; neither supplies a reading of the event in design-basis terms.

The event is the coordination class's reach case and the event of its first consequence chain, a common act (§3.2), and the class cites it for the isolation of the management channel (§7.3, CDB-4), a requirement it states at §4.1. Most of what follows confirms the class's reading in direction. The findings that go further are about where the reach came from, what the management channel shared with the traffic it managed, what it took to stop the storm once its source was removed, what the operations center could see while it lasted, and whose 911 traffic rode the network. This instrument is the home of the finding that a center which cannot see its network has lost it (F-4), which DBA-COMM-RE-8 (*The Rogers Outage, July 2022*) F-5 cites.

1.2 Position and Scope

This is a reference event instrument in the sense of the series convention: it reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for the class and framework to disposition. It was issued under the Communications Sector Framework v0.12, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 at v0.11 on the proposal of *Operations and Coordination*.

The scope is the outage from its first symptoms at 3:40 a.m. Eastern Standard Time on 27 December 2018 to the carrier's determination at 12:01 p.m. on 29 December that the backbone had stabilized, read through the cause, the propagation, the loss of remote access, the restoration, the effect on 911 and the notification of 911 centers. The Commission's consent decree, the Washington penalty and the later rulemaking that cites the event are context. Times are Eastern Standard Time, as the Commission's report gives them. Two matters are set on the author's decisions of 7 October 2026. The Commission's report makes no finding that the management channel was not isolated from the network it managed; this instrument states the Commission's facts and draws its own reading of isolation from them, as its own (F-2). And where the

Commission's record and the Washington commission's differ, on the cause of that state's 911 failures and on whether the carrier notified the 911 centers it served, the two accounts are stated side by side, each attributed, and no finding is made between them. Officials are named by role.

The primary sector is Communications. The equipment that failed lies within the transport network as DBA-COMM-TR-FC1 (*Transport and Backbone*) defines it, and the reach of the failure is the coordination class's measure, which is why the coordination class proposed it. Its effects crossed into the critical-services overlay (framework §3.7) and into Emergency Services through 911, the impacted sector. The constructs applied are defined in the Foundational Definitions and cited, not restated: FD-BL (*The Bright Line*), FD-BR (*The Broker*), FD-LD (*Layer Decomposition and Determinism*), FD-LS (*Local Sovereignty*), FD-BL-D1 (*Mode of Discharge*) and FD-BL-D2 (*Indication Integrity*). FD-REG holds their versions.

2. Sources and Attribution

2.1 Sources and Their Limit

The principal source is the report of the Commission's Public Safety and Homeland Security Bureau of 19 August 2019 on the outage, read in full. The others are the Commission's Order and Consent Decree of 14 December 2020; the carrier's reason-for-outage letter to customers of 31 December 2018, posted publicly by a customer, which is the carrier's own account; the Washington commission's Final Order 08 of 9 June 2023 and Order 10 of 13 November 2023 in Docket UT-181051; the Bureau's public notice of 15 October 2019; the Commission's rulemaking of April 2021 and order of November 2022 on notification of 911 outages; and the reporting of Light Reading, NBC News and the Associated Press, including the carrier's chief executive's remarks of January 2019. No non-public material is used, and no statement by the equipment vendor and no result of its forensic investigation has been found.

The structural facts are established and not contested: a switching module in the carrier's Denver node generated four malformed management packets; they passed the filter of a management channel built into the vendor's nodes, which was enabled by default and which the carrier had neither configured nor used; each node rebroadcast them to every node connected to it, without end; the load disrupted the nodes' ability to carry traffic and to be reached remotely; and the outage lasted almost thirty-seven hours on the Commission's figure. The carrier's account is its own, and where it differs from the Commission's report the report is preferred. Neither the carrier nor the vendor found how or why the module generated the packets, and this instrument does not infer it.

2.2 Two Regulators, Two Accounts

The Commission frames the packet storm as the cause of the outage, which it calls "an equipment failure catastrophically exacerbated by a network configuration error". The Washington commission held that the packet storm itself was not the proximate cause of the 911

failures in that state. It found the violation in the carrier's failure, during the transition of the state's 911 service to Comtech, to take reasonable steps to see that the interconnection with Comtech was built and configured properly, including whether Comtech's circuits, all obtained from the carrier, were diverse; those circuits all failed, while few if any calls to the centers the carrier still served directly failed. Both accounts are stated as their sources', and this instrument makes no finding between them. The two do not conflict on what happened in the network; they differ on where responsibility for the loss of 911 lies.

On notification, the Commission's report records the carrier's statement that it timely notified the 911 centers it served and was in contact with them throughout. That is the carrier's assertion as the report records it; the Bureau made no finding on it. The Washington commission found that the carrier notified neither the commission nor any of the fifteen 911 centers in the state it still served, and penalized it for each of the fifteen. This instrument states both and makes no finding between them.

3. The Event

CenturyLink operated six separately managed long-haul networks. The one that failed provided high-speed data transport over optical fiber and carried 911, voice over internet protocol, local and long-distance voice, Ethernet, the carrier's internet backbone and other services, for the carrier and for other providers. It used nodes supplied by Infinera. Within each node a switching module moved packets from inbound to outbound line modules, and the nodes possessed a proprietary internode management channel, designed to let line modules send packets directly to other connected nodes for very fast rerouting around a failure. To keep management instructions off it, the channel carried a filter that blocked packets of sixty-four bytes or fewer, the size of the nodes' management packets. The vendor supplied the channel enabled by default; the carrier was aware of it but neither configured nor used it.

On the Commission's account, a switching module in the Denver node spontaneously generated four malformed management packets, each with a broadcast destination address, a valid header and checksum, no expiration time, and a size larger than sixty-four bytes. Because they were larger than sixty-four bytes the filter did not stop them, and they entered the enabled and unconfigured management channel. The carrier's letter calls the failed component a faulty network management card; where the two differ, the Commission's report is preferred. Because of their broadcast address, the packets were delivered to every connected node, and each node that received one retransmitted it to all its connected nodes, including the node where the packets originated, indefinitely. The exponentially growing traffic consumed the nodes' processing, which disrupted their internal synchronization, and without it the nodes could no longer route and transmit data. The storm did not trigger any processor-utilization alarm.

Symptoms were present at 3:40 a.m. on 27 December, on the carrier's analysis, and it learned of the outage at 3:56 a.m. from a customer near New Orleans. Its administrators could not connect to the nodes remotely, because the nodes were overloaded, and the carrier's letter says the event

impaired its visibility into the management network and prolonged the outage. Engineers were sent to Omaha and Kansas City to log in to affected nodes directly, and in Kansas City a packet capture pointed to the Denver node. At 9:02 p.m. engineers identified and removed the Denver module, but the packets continued to replicate and transit the network. At 12:09 a.m. on 28 December a network engineer began instructing nodes no longer to acknowledge the malformed packets, and the management channel was disabled. At 11:30 a.m. every node was again reachable remotely, at 11:36 p.m. all nodes had been restored, and at 12:01 p.m. on 29 December the carrier determined that the backbone had stabilized.

As many as twenty-two million customers in thirty-nine states were affected, including about seventeen million in twenty-nine states who lacked reliable access to 911, and at least 886 calls to 911 were not delivered; the Commission was not aware of any harm to life or property. Much of the loss was in other providers' services that rode the network. Comtech, which processed 911 calls for Washington, north central Texas and Tier 3 wireless carriers, depended on signaling links all four of whose diverse paths rode the carrier's transport. Verizon Wireless, on its older CDMA network, AT&T, Comcast's voice service, Cox and several regional carriers reported losses, and the 911 providers for Massachusetts and for parts of Texas and Montana reported failed calls. In Washington the state commission found that more than 13,000 calls, about half of those sent to centers served through Comtech, failed.

The carrier disabled the channel across the network and applied filtering, and the vendor disabled it on new nodes and updated its manual to recommend disabling it where unused. The Commission's report drew four lessons: features not in use should be turned off; catch-all filters should allow only expected traffic; monitoring should include processor and memory alarms that are regularly audited; and procedures for network repair should address cases in which normal monitoring is unavailable. It said those practices could have prevented the outage, or at least mitigated its effects. The Commission settled its investigation in December 2020 for $500,000, without a legal finding, and the Washington commission imposed $1,315,000 in June 2023, for 13,000 failed calls and for the fifteen centers not notified.

4. Technical Decomposition

The event decomposes into initiating elements (I-n), which made the outage possible, and effect elements (E-n), which carried it to the network's users.

I-1: a management channel that was the vendor's, supplied enabled, and never configured or used by the carrier, a path by which any line module could reach every connected node. I-2: the channel's filter, which stopped packets of sixty-four bytes or fewer regardless of content, so that larger packets with a broadcast address and no expiration passed. I-3: one switching module in Denver that generated four malformed packets, for a reason never found. E-1: a storm with the network's reach, every node rebroadcasting the packets to every connected node without end, the load in each node's processing stopping it carrying traffic, and removing the source not stopping the storm. E-2: the operations center blind, the overloaded nodes unreachable remotely,

engineers sent to the nodes, and the network not fully reachable again for about thirty-two hours. E-3: other providers' 911, the network carrying 911 and other traffic for other providers, including all four diverse signaling paths of a 911 processor. E-4: notification contested, the carrier saying it notified the 911 centers it served and the Washington commission finding it notified none of fifteen.

Composed, the elements describe a network in which one fault in one module reached every node, through a channel the operator did not use, and took down the network's ability to carry traffic and the operator's ability to see it in the same act; a storm that outlived its source; a restoration done by engineers at the nodes; and a loss of 911 that fell mostly on other providers' customers. Their composition is what *Operations and Coordination* calls chain C, a common act (§3.2), at the reach of a national transport network, without an act at the operations center at all.

Table 4-1. Event elements and their mapping. Section references are to Operations and Coordination v0.2 unless another instrument is named.

Ref	Description	Provisions engaged	Finding
I-1	Vendor management channel enabled by default; neither configured nor used by the carrier	§3.1, §4.1; Transport and Backbone §3.2, §4.5	F-1
I-2	Filter blocked only packets of 64 bytes or fewer, regardless of content	§4.1; Transport and Backbone §4.5	F-1
I-3	Denver switching module generated four malformed packets; cause never found	§4.1; §3.2	F-1
E-1	Rebroadcast without end; node processing consumed; forwarding lost; storm outlived its source	§4.1, §6.3; Transport and Backbone §7.2	F-2, F-3
E-2	Nodes unreachable remotely; engineers on site; all nodes reachable again at 11:30 a.m. on 28 December	§4.3, §5.4, §7.3	F-4
E-3	17 million in 29 states without reliable 911; a 911 processor's four diverse links on one network	§3.1, §10.1; Transport and Backbone §2.1, §3.2; framework §3.7	F-5
E-4	Carrier's statement of timely notice; Washington finding that none of 15 centers was notified	§4.8	F-6

5. Tier Determination

Against the four characteristics of military action, as *Operations and Coordination* §5.2 states them for coordination: intentionality is absent, the packets having been generated spontaneously, for a reason the carrier and the vendor could not find, and the carrier's letter saying the outage was not caused by a source external to its network, such as a security incident, with nothing in the record suggesting otherwise; persistence is absent, no actor having been present in the network; precision is absent, the fault reaching what the channel reached and falling two days after Christmas by chance; and denial is present in effect, not in design, transport and with it 911 for some seventeen million customers having been denied for more than a day, and the operator

denied remote access to its own network by the same storm, while nothing denied the means of restoration.

The event does not meet the threshold for the military-action extension. It is a non-adversarial event nearest the class's first design-basis category, a wrong act or tool (§4.1), though it arose in the equipment rather than at the operations center that §4.1 postulates, with the loss of the center's sight of its network (§4.3) realized in its restoration, in chain C, at the reach of a national transport network. The provisional placement is confirmed. That is the tier determination returned for the DBA Reference Event Register.

What a directed actor able to place such a packet on such a channel could do (choose a disaster or a military crisis as the moment, place packets at several nodes so that removing one source does nothing, and count on the same storm to blind the operator so that restoration must be done by people at the nodes when roads and power are least available) runs forward from the event rather than back through it. It is offered to Project 9 as material for the class's governing military-action scenario and is not developed here. Its bearing on this reading is recorded in Findings F-1 to F-4.

6. Characterization Against the Design Basis

Section references are to *Operations and Coordination* v0.2, the version this instrument read, and to *Transport and Backbone* v0.1 where the equipment is that class's.

6.1 §3.1 and §4.1 Where the Reach Came From

Missed. The class counts authority reach as how much of the network one act at an operations center, or one compromised operations account or tool, can change at once (§3.1), and its first design-basis event is a wrong act or tool at the center carried to many elements (§4.1). At CenturyLink there was no act at the center. The reach came from a function built into every node, supplied enabled and left unused, by which one line module could reach every connected node without management instructions. *Transport and Backbone* already counts a management system as an element shared by every path it controls (*Transport and Backbone* §3.2, §4.5), and credits paths as independent only when they share no control system, codebase or vendor, but it does not name a dormant function in the equipment itself as an element by which one node reaches every other. A channel no one uses is still a channel, and its reach is the reach of whatever can put a packet on it. Finding F-1 records it.

6.2 §4.1 and Transport and Backbone §7.2 What the Channel Shared

Missed, on this instrument's reading. The class requires the management channel to be isolated from the network it manages, so that a fault in one does not travel through the other (§4.1), and *Transport and Backbone* finds that management rides the fiber it manages and counts a shared path as one path (*Transport and Backbone* §7.2). Both count isolation by path. The Commission's report does not say whether the channel shared fiber or wavelengths with

customer traffic. What the report does establish is that the management packets consumed the processing on which each node's forwarding of traffic depended, and that the same overload cut off remote access. That is a sharing the class does not count: management and traffic shared the node's processing, and the channel by which the operator could have stopped the storm failed with the traffic. This is this instrument's reading of the Commission's facts, not a finding of the Commission. Finding F-2 records it.

6.3 §4.1 and §6.3 Stopping What Had Started

Held, with a gap. The class requires that the management channel be able to stop what it has started (§4.1), and that acts of wide reach be authorized in stages (§6.3). At CenturyLink the source was found and removed seventeen hours in, and the storm went on, because every node was now a source. It stopped only when an engineer began instructing nodes not to acknowledge the packets and the channel was disabled, three hours later. The act that ended the storm was itself an act of the whole network's reach, taken during restoration, and the record names only the role of the person who took it. A storm that sustains itself is not stopped by removing what started it; the means of stopping it must reach every node by a path the storm does not take, and the act is one the restoration plan stages in advance, as the class requires of isolation (§6.3). Finding F-3 records it.

6.4 §4.3, §5.4 and §7.3 The Operations Center Blind

Held, with a gap. The class postulates the loss of the operations center (§4.3), and requires that each part of the network keep serving without it for the stand-alone time (§5.4) and that field crews be able to carry out restoration without direction from the center (§7.3, CDB-3). At CenturyLink the center stood and received alarms, but could not connect to its nodes; the carrier's engineers restored it by logging in at the nodes and taking packet captures there, which is the field-executable restoration CDB-3 asks for, though the record shows it done at the center's direction rather than without it. The class does not name this case. Loss of the center's sight of its network, while the center itself is intact, is the loss of the center in effect, and the stand-alone time is measured against it: here not every node was again reachable remotely until about thirty-two hours after the first symptoms. The Commission's fourth lesson, repair procedures for when normal monitoring is unavailable, is the same requirement. Finding F-4 records it.

6.5 §3.1, §10.1 and Transport and Backbone §2.1 Other Providers' 911

Held, with a gap. The class counts authority reach in network elements and in the subscribers and dependents those elements serve, and leaves the method for counting dependents to be developed (§3.1, §10.1). Most of the loss of 911 was not in the carrier's own service: it was in a 911 processor whose four diverse signaling links all rode the carrier's network, in Verizon Wireless's older network, in Comcast's voice service and in the 911 providers of other states. *Transport and Backbone* names apparent diversity that is not real as the commonest cause of large transport failures (*Transport and Backbone* §2.1); the Washington commission found the violation in the

carrier's failure to take reasonable steps over an interconnection whose circuits all came from the carrier. A carrier's authority reach includes the 911 paths of others that ride its network, and the paths of one 911 service counted as diverse are one path if they ride one transport network. Finding F-5 records it.

6.6 §4.8 Notification

Held, on a contested record. The class requires the carrier to notify the 911 centers an outage affects within a stated time, over a path independent of the failed network, triggered by the detection that finds the failure, and recorded (§4.8). Here the carrier said it notified the centers it served and the Washington commission found it notified none of fifteen. The record cannot decide between them, and that is the point of the requirement that notice be recorded. Finding F-6 records it.

6.7 §6.6 The AI-Governance Design Basis

Not exercised. Nothing in the record shows a component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the outage or the restoration, and the event does not test the class's AI provisions.

6.8 Summary of Design Basis Performance

Of the seven readings, notification Held, on a contested record; stopping what had started, the blind center and other providers' 911 Held, with a gap; where the reach came from and what the channel shared were Missed; and the AI-governance design basis was Not exercised. The class's common-act chain, its requirement of a management channel isolated from what it manages and able to stop what it starts, its field-executable procedures and its notification requirement all describe what happened or what was missing. Where it falls short, it does so on what the event added: reach without an act at the center, the sharing of the node's processing, a storm that outlived its source, a center blind but intact, and other providers' 911 on one network.

7. Findings

Section references are to *Operations and Coordination* v0.2 unless another instrument is named.

7.1 F-1 — An Unused Function in the Equipment Counts in Reach

The management channel that carried the storm was the vendor's, supplied enabled and never configured or used by the carrier, and its filter screened packets by size alone. The class counts the reach of acts at the center and of compromised tools (§3.1, §4.1). The finding is that the class should count in authority reach every function by which one network element can reach others without a governed act, including functions supplied enabled and not in use; that a function not in use is disabled, and a function in use admits only the traffic it expects, as the Commission's first two lessons state; and that *Transport and Backbone* should count such a function, like a

management system, as an element shared by every path it reaches. Affected: §3.1, §4.1; *Transport and Backbone* §3.2, §4.5. Disposition: R-1.

7.2 F-2 — Isolation Is Counted in Processing as Well as Path

The management packets consumed the processing on which each node's forwarding of traffic depended, and the same overload cut off the operator's remote access. The Commission's report makes no finding on isolation; this is this instrument's reading of its facts. The class and *Transport and Backbone* count isolation of management by path (§4.1; *Transport and Backbone* §7.2). The finding is that isolation of the management channel is counted in the resources of each element as well as in its path: management traffic cannot consume the processing or memory on which forwarding and the operator's access depend, and processor and memory alarms, regularly audited, are part of the element's protection, as the Commission's third lesson states. Affected: §4.1, §7.3 (CDB-4); *Transport and Backbone* §7.2, §7.3. Disposition: R-2.

7.3 F-3 — A Storm That Sustains Itself Is Stopped at Every Node

The source was removed seventeen hours in, and the storm went on until nodes were instructed not to acknowledge the packets and the channel was disabled, three hours later. The class requires that the management channel be able to stop what it has started (§4.1). The finding is that the class should state that the means of stopping a self-sustaining storm reach every element by a path the storm does not take; that the act of stopping it, which has the whole network's reach, is staged in the restoration plan and authorized at the time by a qualified holder, as isolation is (§6.3); and that the record of the act names the holder, which this record does not. Affected: §4.1, §6.3; *Core Switching, Routing and Signaling* §4.3. Disposition: R-3.

7.4 F-4 — A Center That Cannot See Its Network Has Lost It

The carrier could not reach its nodes remotely in the first hours of the outage, and not every node was reachable again for about thirty-two hours, and restored the network by sending engineers to them. The class postulates the loss of the center (§4.3) and requires field-executable procedures (§7.3, CDB-3). The finding confirms CDB-3 and is that the class should state that the loss of the center includes the loss of its sight and reach into its network while the center stands, that stand-alone time is counted against that case too, and that restoration procedures are written for it, as the Commission's fourth lesson states. Under the triage of 9 October 2026 this instrument is the home of that finding for the series; DBA-COMM-RE-8 F-5 cites it for the Rogers restorers who could not see the failed routers' logs, and keeps visibility on the independent path as its own. Affected: §4.3, §5.4, §7.3 (CDB-3, CDB-7). Disposition: R-4.

7.5 F-5 — Other Providers' 911 Counts in Reach

Some seventeen million customers in twenty-nine states lacked reliable 911, mostly through other providers whose 911 paths rode the carrier's network, including a 911 processor all four of

whose diverse signaling links did. The Washington commission held that the storm itself was not the proximate cause of that state's failures and found the violation in the carrier's failure to take reasonable steps over the non-diverse interconnection with Comtech. The class leaves the method for counting dependents open (§3.1, §10.1). The finding is that the method counts, among a carrier's dependents, the 911 and priority paths of other providers that ride its network; that *Transport and Backbone* counts paths of one 911 service as diverse only when they ride different transport networks; and that the dependence of a 911 service on one transport network is counted under the elevation rule of the critical-services overlay. Affected: §3.1, §10.1; *Transport and Backbone* §2.1, §3.2, §8.1; framework §3.7, §6.1. Disposition: R-5.

7.6 F-6 — A Notification Not Recorded Cannot Be Shown

The carrier said it notified the 911 centers it served; the Washington commission found it notified none of fifteen; the record cannot decide between them. The finding confirms §4.8, that notification is triggered by detection and recorded, and adds this event to the pattern of late or doubtful notification of 911 centers whose home is *Multistate 911 Outage* F-7, with *Derecho 911 Outages* F-1 and DBA-COMM-RE-7 F-6. Affected: §4.8. Disposition: R-6.

7.7 F-7 — Withdrawn at v0.2

Its substance corrected the class's own account of the event at *Operations and Coordination* §2.3 and §7.3 (the switching module and the vendor's unused channel; twenty-nine states, seventeen million customers and at least 886 failed calls; the isolation reading attributed to this instrument rather than the Commission; the practices in the Commission's words). That is a correction to another instrument rather than a test of the design basis, and the class took it in at its v0.3, so no erratum is owed. Its naming of the impacted sector is carried by R-8.

8. Recommendation Register

Priority is stated on the series' one-to-three scale, where 1 is the most operationally consequential. R-1 to R-6 dispose to *Operations and Coordination*, the transport elements of R-1, R-2 and R-5 also to *Transport and Backbone*, the storm element of R-3 to *Core Switching, Routing and Signaling*, and R-8 to the register.

ID	Recommendation	Affected	Direction	Pri
R-1	Count unused functions in reach	§3.1, §4.1; Transport and Backbone §3.2, §4.5	Count in authority reach every function by which one element reaches others without a governed act, including functions enabled and unused; disable what is unused and filter what is used to expected traffic; count such a function as a shared element.	1
R-2	Isolate management in processing	§4.1, §7.3; Transport and Backbone §7.2, §7.3	State that management traffic cannot consume the resources on which forwarding and the operator's access depend, with processor and memory alarms audited; this instrument's reading, not the	1

			Commission's finding.	
R-3	Stop a storm at every node	§4.1, §6.3; Core Switching, Routing and Signaling §4.3	State that the means of stopping a self-sustaining storm reach every element by a path the storm does not take, the act staged in the restoration plan and authorized at the time, with the holder recorded.	2
R-4	Name the blind center	§4.3, §5.4, §7.3	State that loss of the center includes loss of its sight and reach while it stands, count stand-alone time against it, and write restoration procedures for it.	2
R-5	Count other providers' 911	§3.1, §10.1; Transport and Backbone §2.1, §3.2, §8.1; framework §3.7, §6.1	Count other providers' 911 and priority paths among a carrier's dependents; credit a 911 service's paths as diverse only across transport networks; count the dependence under the elevation rule.	1
R-6	Confirm recorded notification	§4.8	Confirm notification triggered by detection and recorded; record a further event in which notification of 911 centers is late or in doubt.	3
R-7	Withdrawn at v0.2	§2.3, §7.3	Its substance, correcting the class's account of the event, was taken in by Operations and Coordination at its v0.3; withdrawn with F-7.	—
R-8	Name the sector	DBA Reference Event Register	Ask that the register name Emergency Services as an impacted sector. Entered by Project 0.	3

9. Open Items

OI-1, the register row, is closed: Project 0 entered the tier determination of Section 5, Emergency Services as the impacted sector, the instrument version and the counts. OI-2 stands: no forensic result from the vendor has been published, and the Washington commission staff's investigation report, which holds its technical findings and the narratives of failed 911 calls, has not been read; either would bear on Sections 2 and 3 and on F-1, F-2 and F-5. OI-3 is closed: *Operations and Coordination* took in R-1 to R-7 at its v0.3, with its account corrected at §2.3; *Core Switching, Routing and Signaling* took in R-3's storm element at its v0.3; and *Transport and Backbone* took in the transport elements of R-1, R-2 and R-5 at its v0.2. Section 6 remains a reading of the class at v0.2, as issued.

10. Standing of This Document

This instrument establishes where the CenturyLink outage of 27 to 29 December 2018 sits within the Communications analytical structure, which provisions of *Operations and Coordination*, *Transport and Backbone* and the framework bore on it, and what the class did and did not correctly draw from it. It does not modify the class or the framework. It makes no finding on the cause of the packets, no finding between the two regulators' accounts, and no finding of law.

CenturyLink is the coordination class's reach case: what one fault does with the reach of a whole network and no actor behind it. The Salt Typhoon intrusion (DBA-COMM-RE-14) is the commons case, and national roaming in Ukraine (DBA-COMM-RE-15) the success case. The core class's events show the same pattern at the core: at AT&T one change reached the whole mobile network in three minutes (DBA-COMM-RE-7), and at Rogers the restorers could not reach equipment because the management network depended on the core that had failed (DBA-COMM-RE-8); CenturyLink adds a reach that no one had made and a blindness the storm itself caused. The instrument should be revised on any revision of *Operations and Coordination* or *Transport and Backbone* that dispositions these findings, or on the material of OI-2.

References

Federal Communications Commission, Public Safety and Homeland Security Bureau, *December 27, 2018 CenturyLink Network Outage Report*, 19 August 2019; Enforcement Bureau, *In the Matter of CenturyLink, Inc.*, Order and Consent Decree, DA 20-1469, 14 December 2020; Public Safety and Homeland Security Bureau, Public Notice DA 19-1039, 15 October 2019; Third Notice of Proposed Rulemaking, FCC 21-45, April 2021, and Second Report and Order, FCC 22-88, November 2022. Washington Utilities and Transportation Commission, Docket UT-181051, Final Order 08, 9 June 2023, and Order 10, 13 November 2023. CenturyLink, customer reason-for-outage letter of 31 December 2018, as posted by a customer. Light Reading, 2 and 10 January 2019; NBC News, 28 December 2018; the *Salt Lake Tribune* (Associated Press), 28 December 2018.

DBA instruments: the Communications Sector Framework, under which this instrument was issued at v0.12; *Operations and Coordination* (DBA-COMM-OpsCo), the proposing class, read at v0.2; *Transport and Backbone* (DBA-COMM-TR-FC1) and *Core Switching, Routing and Signaling* (DBA-COMM-CR-FC3); *The AT&T Mobility Outage, February 2024* (DBA-COMM-RE-7), *The Rogers Outage, July 2022* (DBA-COMM-RE-8), *The Salt Typhoon Intrusion, Disclosed 2024* (DBA-COMM-RE-14) and *National Roaming in Ukraine, 2022 to 2026* (DBA-COMM-RE-15); *The 911 Outages After the Derecho of June 2012* (DBA-ESS-RE-1) and *The Multistate 911 Outage of April 2014* (DBA-ESS-RE-2), cited by the short titles *Derecho 911 Outages* and *Multistate 911 Outage*; the Military Action Communications Design Basis (DBA-MA-CDB); and the DBA-MA Parent Framework, source of the four military-action characteristics.

Revision History

Version 0.1, October 2026: issued in Project 9 under the Communications Sector Framework v0.12, for the event conferred as DBA-COMM-RE-13 on the proposal of *Operations and Coordination*. The outage from its first symptoms on 27 December 2018 to the backbone's stabilization on 29 December, the enforcement and later rulemaking as context. On the author's decisions of 7 October 2026: the Commission's facts stated and the reading of isolation drawn as

this instrument's own, not the Commission's; the Commission's and the Washington commission's accounts of the cause of that state's 911 failures and of notification stated side by side, with no finding between them. Officials named by role; tier set provisionally and confirmed. An independent check corrected the record before issue. Tier determination: non-adversarial event nearest the class's first design-basis category, arising in the equipment, chain C. Findings F-1 to F-7; recommendations R-1 to R-8.

Version 0.2, 10 October 2026: migrated to Project 11 under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), compressed under the Reference Event Compression Standard v0.3 and brought to house style. Cut: the general description of the reference event instrument; the restatement of the governing instruments (now citations); the evidentiary-class and currency sections, merged into Section 2.1; the event narrative, cut to the facts the findings rest on, with the Washington commission's proceedings trimmed to the findings and penalty the instrument relies on; the decomposition and the four characteristics, tightened; the constructed analog of Section 5.6, which runs forward and is offered to Project 9 as use-case material; the open items, relationship and maintenance sections, merged into Sections 9 and 10; and the references, condensed. Section 6 now uses the standard's verdicts, and the separate account-of-the-event subsection is removed with F-7. Under the triage of 9 October 2026 this instrument is the home of the finding that a center which cannot see its network has lost it (F-4), which DBA-COMM-RE-8 F-5 cites at its v0.2; no finding moved here. F-6 is cut to a cross-reference to the notification pattern's home at *Multistate 911 Outage* F-7; R-6 is unchanged. Stubs created: F-7 and R-7, whose correction of *Operations and Coordination* §2.3 and §7.3 the class had already taken in at its v0.3, so that no erratum is owed, with F-7's naming of the sector carried by R-8. Unchanged: the designator, title, tier determination, attribution statements, sources, and the numbering of F-1 to F-6, R-1 to R-6 and R-8, whose substance stands. OI-1 closed on the register entry, and OI-3 on the classes' revisions. Verified before migration: every finding and recommendation number present at v0.1 is present here as an entry or a stub.